

$$2 = \frac{1+3}{2} \geq \sqrt{1 \cdot 3} \quad 4 = \frac{3+5}{2} \geq \sqrt{3 \cdot 5}$$

$$0 < \text{原式} \leq \frac{1}{\sqrt{1 \cdot 3}} \cdot \frac{3}{\sqrt{3 \cdot 5}} \cdot \frac{5}{\sqrt{5 \cdot 7}} \cdots \frac{2n-1}{\sqrt{(2n-1) \cdot 2n+1}} = \frac{1}{\sqrt{2n+1}} \rightarrow 0$$

$$\lim_{n \rightarrow \infty} \frac{1}{2} \cdot \frac{3}{4} \cdots \frac{2n-1}{2n} = 0$$

$$\text{Eg2: } \lim_{n \rightarrow \infty} \frac{\sum_{p=1}^n p!}{n!} = 1$$

$$1 < \frac{\sum_{p=1}^n p!}{n!} = \frac{1! + 2! + \cdots + n!}{n!} = 1 + \frac{1}{n} + \frac{1}{n(n-1)} + \cdots + \frac{1}{n!}$$

$$< 1 + \frac{1}{n} + \frac{n-2}{n(n-1)} \rightarrow 1$$

$$\text{Eg3: } \lim_{n \rightarrow \infty} [(n+1)^a - n^a] = 0, 0 < a < 1$$

$$0 < (n+1)^a - n^a < n^a [1 + \frac{1}{n}]^a - 1 < n^{\frac{1}{1-a}} \rightarrow 0$$

② 单调有界数列必有极限

(1) 单调增有上界的数列必有极限

$$x_1 \leq x_2 \leq \cdots \leq x_n \leq x_{n+1} \leq \cdots \leq M \Rightarrow \lim_{n \rightarrow \infty} x_n = \xi = \sup \{x_n\}$$

上确界, 最小上界
 ξ 上界 + $\forall \varepsilon, \exists x_0 \in A$
 $x_0 > \xi - \varepsilon$
 ξ 确界原理
 一个集合若有上下界
 必有上下确界

(2) 单调减有下界的数列必有极限

$$x_1 \geq x_2 \geq \cdots \geq x_n \geq x_{n+1} \geq \cdots \geq M$$

$$\Rightarrow \lim_{n \rightarrow \infty} x_n = \eta = \inf \{x_n\}$$

证明: 不妨设 $\{x_n\}$ 单调增, 有上界,
由确界原理, 存在 $\sup \{x_n\} = \xi$

$$\forall \varepsilon > 0, \exists x_{n_0} > \xi - \varepsilon$$

$$\text{当 } n > n_0 \text{ 时 } x_n \geq x_{n_0} > \xi - \varepsilon$$

$$\text{要证 } \forall \varepsilon > 0, \exists N \quad \forall n > N \quad |x_n - \xi| < \varepsilon$$

$$\text{即证: } x_n > \xi - \varepsilon \quad \text{显然得证}$$

Eg1: 设 $x_n = (1 + \frac{1}{n})^n$ ($n=1, 2, \dots$) 证明数列 $\{x_n\}$ 极限存在

证: 先证单调性 1° 均值不等式 2° $x_{n+1} - x_n > 0$ 3° $\frac{x_{n+1}}{x_n} \geq 1$ 4° 归纳递推

$$x_n = (1 + \frac{1}{n}) (1 + \frac{1}{n}) \cdots (1 + \frac{1}{n}) \leq \left[\frac{1+n(1+\frac{1}{n})}{n+1} \right]^{n+1} = (1 + \frac{1}{n+1})^{n+1} = x_{n+1}$$

再证有界性

$$\begin{aligned} x_n &= (1 + \frac{1}{n})^n = 1 + 1 + C_n^2 \cdot \frac{1}{n^2} + C_n^3 \cdot \frac{1}{n^3} + \cdots + C_n^n \cdot \frac{1}{n^n} \\ &= 2 + \frac{n!}{n^2 2! (n-2)!} + \frac{n!}{n^3 3! (n-3)!} + \cdots + \frac{1}{n!} \\ &< 2 + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{n!} \\ &< 1 + (\frac{1}{2})^0 + (\frac{1}{2})^1 + \cdots + (\frac{1}{2})^{n-1} = 3 - (\frac{1}{2})^{n-1} < 3 \end{aligned}$$

$$\lim_{n \rightarrow \infty} (1 + \frac{1}{n})^n = e$$

Eg2: 证明数列 $\sqrt{2}, \sqrt{2+\sqrt{2}}, \dots, \sqrt{2+\sqrt{2+\cdots+\sqrt{2}}} \cdots$ 收敛, 并求其极限

证: 显然单增 且 $x_{n+1} = \sqrt{2+x_n}$ 开根号

$x_1 = \sqrt{2} < 2, x_2 = \sqrt{2+\sqrt{2}} < 2$ 由 $x_n < 2$ 得 $x_{n+1} < 2$
∴ 有上界

③ 柯西收敛原理

$\{x_n\}$ 收敛 $\Leftrightarrow \forall \varepsilon > 0, \exists N \in \mathbb{N}^+, \forall m, n > N, |x_m - x_n| < \varepsilon$

函数的极限

1. ε - X 定义: $\lim_{x \rightarrow \infty} f(x) = A \Leftrightarrow$

$\forall \varepsilon > 0, \exists X > 0, \text{当 } |x| > X, \text{有 } |f(x) - A| < \varepsilon$

定义: 如果 $\lim_{x \rightarrow \infty} f(x) = A$, 则称直线 $y = A$ 为函数 $y = f(x)$ 的图像的水平渐近线

$$\lim_{x \rightarrow \infty} f(x) = A \Leftrightarrow \lim_{x \rightarrow +\infty} f(x) = \lim_{x \rightarrow -\infty} f(x) = A$$

$$\lim_{x \rightarrow \infty} f(x) \neq A \Leftrightarrow \exists \varepsilon > 0, \forall X > 0, \exists |x_0| > X \text{ 有 } |f(x_0) - A| \geq \varepsilon$$

2. 自变量趋向有限值时函数的极限

(1) 双侧极限

$0 < |x - x_0| < \delta$ 表示 $x \rightarrow x_0$ 的过程

$$U^o(x_0, \delta)$$

$$0 \leq |x - x_0| < \delta \quad U(x_0, \delta) \quad U(x_0)$$

ε - δ 定义: $\lim_{x \rightarrow x_0} f(x) = A \Leftrightarrow$

$\forall \varepsilon > 0, \exists \delta > 0, \text{当 } 0 < |x - x_0| < \delta \text{ 时有 } |f(x) - A| < \varepsilon$

$$\lim_{x \rightarrow x_0} f(x) \neq A \Leftrightarrow$$

$\exists \varepsilon_0 > 0, \forall \delta > 0, \text{当 } 0 < |x' - x_0| < \delta \text{ 时有 } |f(x') - A| \geq \varepsilon_0$

Eg: 证明 $\lim_{x \rightarrow 2} x^2 = 4$

证明: $\forall \varepsilon$, 要找 δ , 当 $0 < |x-2| < \delta$ 时, 有 $|x^2-4| < \varepsilon$

$$|x+2||x-2| < \varepsilon$$

取 $0 < |x-2| < 1$ 则 $x \in (1, 2) \cup (2, 3)$

$$|x+2| \in (3, 4) \cup (4, 5)$$

$$|x+2||x-2| < 5|x-2| < \varepsilon \Rightarrow |x-2| < \frac{\varepsilon}{5}$$

$$\delta = \min \left\{ 1, \frac{\varepsilon}{5} \right\}$$

(2) 单侧极限

左极限: $\lim_{\substack{x \rightarrow x_0-0 \\ (x \rightarrow x_0^-)}} f(x) = A$ 或 $f(x_0-0) = A \Leftrightarrow$

$\forall \varepsilon > 0, \exists \delta > 0$, 使当 $x_0 - \delta < x < x_0$ 时 恒有 $|f(x) - A| < \varepsilon$

右极限: $\lim_{\substack{x \rightarrow x_0+0 \\ (x \rightarrow x_0^+)}} f(x) = A$ 或 $f(x_0+0) = A \Leftrightarrow$

$\forall \varepsilon > 0, \exists \delta > 0$, 使当 $x_0 < x < x_0 + \delta$ 时 恒有 $|f(x) - A| < \varepsilon$

极限存在定理:

$$\lim_{x \rightarrow x_0} f(x) = A \Leftrightarrow f(x_0-0) = f(x_0+0) = A$$

极限形式	度量	过程描述	目标不等式
$\lim_{n \rightarrow \infty} x_n = a$	$\forall \varepsilon > 0$	$\exists N > 0$, 当 $n > N$ 时	$ x_n - a < \varepsilon$
$\lim_{x \rightarrow \infty} f(x) = a$		$\exists X > 0$, 当 $ x > X$ 时	$ f(x) - a < \varepsilon$
$\lim_{x \rightarrow +\infty} f(x) = a$		$\exists X > 0$, 当 $x > X$ 时	
$\lim_{x \rightarrow -\infty} f(x) = a$		$\exists X > 0$, 当 $x < -X$ 时	
$\lim_{x \rightarrow x_0} f(x) = a$		$\exists \delta > 0$, 当 $0 < x - x_0 < \delta$ 时	
$\lim_{x \rightarrow x_0^+} f(x) = a$		$\exists \delta > 0$, 当 $0 < x - x_0 < \delta$ 时	
$\lim_{x \rightarrow x_0^-} f(x) = a$		$\delta > 0$, 当 $-\delta < x - x_0 < 0$ 时	